

ANIL SEZGIN

Senior AOSP/AAOS & BSP Developer

anil.sezgin@outlook.com · anilsezgin.dev · linkedin.com/in/anilsezgin · Gothenburg, Sweden

SUMMARY

Senior software engineer with over a decade of experience in consumer electronics, defense industry, and automotive. Expert in Embedded Android (AOSP/AAOS), Board Support Package (BSP) development, and QNX. Proven track record in leading platform-level software development across smartphones, tablets, defense systems, and next-generation automotive infotainment platforms.

EXPERIENCE

Lead Software Developer

HaleyTek AB — BSP

May 2023 — Present

Göteborg, Sverige

- ▶ Lead Software Developer Nov 2024 — Present
- ▶ Senior Software Developer May 2023 — Oct 2024

- › Board bring-up of Volvo Cars IVI systems on Qualcomm SA8155 and SA8255 SoCs.
- › Integration of Qualcomm baselines and AOSP/AAOS uplifts from Google manifests.
- › Python automation framework for baseline integrations and AOSP uplifts, with unit and integration tests.
- › Boot time optimizations on QNX Hypervisor and Android VM.
- › Debugging platform-level issues across QNX Hypervisor and Android VM.
- › Addressing CTS/VTS failures on the platform.
- › Secureboot enablement and automation framework development.
- › UFS reprovisioning, partition layout optimization, and QDL tooling for CI/CD.
- › Virtual A/B and dynamic partitions adaptation for software download processes.
- › Incremental OTA implementation for software download processes.

Senior Software Engineer

Aptiv — Advanced Safety / UX

Nov 2021 — Apr 2023

Kraków, Polska

- › Development and maintenance of software download applications wrapping Android's Update Engine.
- › Flashing tool implementation and maintenance wrapping fastboot and QDL
- › Introducing Dynamic partitions and Virtual A/B to customer-specific software download processes.
- › Test implementations for software download processes with Pytest and Behave
- › Automation tool development for software download package creation per build wrapping Google's otatools.
- › Debugging platform-level issues on both QNX and Android Virtual Machine.

Senior Researcher

TÜBİTAK SAGE — Embedded Systems

Oct 2019 — Sep 2021

Ankara, Türkiye

- › Extension and development of ExoPlayer library for real-time video streaming for cockpit HMI
- › Development of video player applications using GStreamer SDK, H.264 over RTP/RTSP for cameras running on Yocto
- › C++ driver developments on CPU cores of Xilinx Zynq 7000 for custom IPs on FPGA for image processing algorithms

Software Architect

Vestel Electronics — Mobile Devices

Dec 2013 — Sep 2019

Manisa, Türkiye

- ▶ Software Architect Dec 2017 — Sep 2019
- ▶ Sr. Software Design Specialist Jan 2015 — Dec 2017
- ▶ Software Design Specialist Jan 2014 — Dec 2014
- ▶ Software Design Engineer Dec 2013

- › Led architectural decisions for Android OS customization on Qualcomm-based smartphones and tablets.
- › Design, development and integration of system services for manufacturing and OTA updates.
- › Android OS bring-up on custom boards based on Qualcomm chipsets.
- › Peripheral bring-ups of display (MIPI-DSI), audio, sensors (I2C, SPI) on Android Kernel.
- › Device driver development for touch-screens, EMR sheets/pens and sensors.
- › BSP-level development and integration for tablet and smartphone product lines.
- › Integration of Qualcomm baselines.
- › Average sleep current and charging optimizations for battery.
- › Boot time optimizations of Android OS.
- › Addressing platform-level issues along with CTS/VTS compliance.
- › GMS and Widevine DRM integrations.

EDUCATION

M.S. in Computer Engineering

2017 — 2019

Özyeğin University

B.S. in Electrical & Electronics Eng.

2008 — 2013

Middle East Technical University

AREAS OF EXPERTISE

Embedded Android (AOSP)
Android Automotive (AAOS) BSP Development
Linux Device Drivers QNX Hypervisor
OTA / SWDL Boot Optimization
CTS/VTS Compliance Secureboot
CI/CD Automation GStreamer / Multimedia
Qualcomm SoCs

PROGRAMMING

C++ 98/11/14/17 C Python 3.x Java
Shell Makefile CMake

TOOLS & PLATFORMS

Android Studio Git Gerrit Claude GDB
ADB Repo Tool VS Code

LANGUAGES

Turkish — Native
English — Fluent Swedish — Beginner

HONORS

65th / 1,600,000 2008
University Entrance Exam (ÖSS)

PROJECTS

Volvo EX90 Infotainment / Volvo EX60 Infotainment / Volvo EX30 Infotainment

HaleyTek AB & Aptiv

Cockpit HMI

TÜBİTAK SAGE

Vestel Venus V5 / Vestel Venus V4 / Vestel Venus V3 / THOR Tablet / V Tab 7 / V Tab 7.85 / V Tab 10.1

Vestel Electronics